

Logbook & Technical Report — CAT

Mobile Application

Module: Mobile Application Development

Project: Nikah Planner — Wedding planning mobile app

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1. Project overview

1.1 Context

Nikah Planner is an Android mobile application that helps couples plan their wedding: budget, guest list, ceremony schedule, vendors, and notifications. This project extends the UI prototype developed earlier in the semester into a full Flutter application.

1.2 CAT objectives

Build a complete mobile application covering concepts from weeks 1 to 5:

CAT requirement	Nikah Planner implementation
User Interface Design	13 Material Design screens, gold/sage theme
Navigation	<code>go_router</code> + 5-tab bottom navigation
Event Handling	Buttons, forms, filters, <code>onTap</code> , <code>onChanged</code>
Local Data Storage	Embedded data (<code>mock_data.dart</code>) + <code>SharedPreferences</code>
Data Retrieval	Data loaded and displayed on each screen
Networking	Firebase Auth + Firestore (cloud)
API Consumption	REST via Firebase SDK
Error Handling	Login/register error messages, form validation

1.3 Technologies

Layer	Technology
Mobile	Flutter 3.x / Dart
Navigation	<code>go_router</code>
State	Riverpod
Local storage	<code>SharedPreferences</code> + embedded data
Cloud backend	Firebase Auth + Cloud Firestore
Web prototype	React + Vite (browser demo)
Version control	Git / GitHub

2. CAT requirements coverage

2.1 User Interface Design

Implementation:

- Animated splash screen (`lib/features/splash/splash_screen.dart`)
- Consistent theme: gold (#B8956A), sage (#8FB5B0), Playfair Display + DM Sans fonts
- Reusable components: `WeddingCard` , `PrimaryButton` , `WeddingInputField`
- 13 responsive screens with `CustomScrollView` , charts (`fl_chart`)

Key files: `lib/core/theme/` , `lib/core/widgets/wedding_widgets.dart`

Instructor mark: _____ / 5

Comments: _____

2.2 Navigation

Implementation:

- `GoRouter` with named routes (`lib/router/app_router.dart`)
- Tab navigation: Home, Budget, Guests, Gallery, More (`MainShell`)
- Stack navigation to sub-pages: Schedule, Vendors, Profile, Settings
- Flow: Splash → Login → Dashboard

Route examples:

```
/splash → /login → /register → /home
/home, /budget, /guests, /gallery, /more
/schedule, /vendors, /profile, /settings, /notifications
```

Instructor mark: _____ / 4

Comments: _____

2.3 Event Handling

Handled events:

- Tap on Login / Register buttons
- Text input via `TextEditingController`
- RSVP and bride/groom side filters (Guest List)

- Event expansion in Schedule (`setState`)
- Notification toggles in Settings
- `onTap` navigation to sub-screens

Files: `login_screen.dart` , `register_screen.dart` , `guest_list_screen.dart` , `schedule_screen.dart`

Instructor mark: _____

Comments: _____

2.4 Local Data Storage

Implementation:

Method	Role	File
Embedded data	Budget, guests, events (demo mode)	<code>lib/core/data/mock_data.dart</code>
SharedPreferences	User accounts and wedding profiles (JSON)	<code>lib/services/local_wedding_store.dart</code>
In-memory session	Active couple data	<code>lib/services/wedding_session.dart</code>

Local storage structure (SharedPreferences):

```

nikah_local_users    → { userId: { email, password, name, weddingId } }
nikah_local_weddings → { weddingId: { full profile JSON } }
nikah_email_index    → { email: userId }

```

Temporary vs permanent storage: see section 12.1

Instructor mark: _____ / 5

Comments: _____

2.5 Data Retrieval

Implementation:

- `MockData` exposes getters read by all screens (budget, guests, events, etc.)
- `WeddingRepository.loadForUser()` loads profile from SharedPreferences or Firestore
- `WeddingSession` provides active data after login

- Guest filters: search + RSVP + bride/groom side

Flow:

Login → loadForUser(userId) → WeddingSession.activate(profile) → MockData reads session → UI displays

Instructor mark: _____ / 4

Comments: _____

2.6 Networking

Implementation:

- **Firestore Authentication:** account creation, email/password sign-in
- **Cloud Firestore:** collections `users/`, `weddings/`, sub-collection `guests/`
- Firebase SDK = async HTTP client to Google Cloud API

Files: `lib/services/auth_service.dart`, `lib/services/firestore_service.dart`

Demo mode (no network): the app works offline with local storage when Firebase is not configured.

Instructor mark (Networking): _____

Comments: _____

2.7 API Consumption

API used: Firebase REST/SDK

Operation	Type	Service
Registration	POST (createUser)	Firebase Auth
Sign in	POST (signIn)	Firebase Auth
Create wedding	POST (set document)	Firestore
Read profile	GET (get document)	Firestore
Guest stream	GET (snapshots)	Firestore

Data format: JSON (Firestore documents, `WeddingProfile.toJson()` serialization)

Instructor mark (API / Networking total): _____ / 5

Comments: _____

2.8 Error Handling

Implementation:

Situation	Handling
Empty login fields	Message: "Please enter email and password"
Password < 8 characters	Message on registration
Email already registered	"This email is already registered"
Account not found	"Account not found. Please register first."
Wrong password	"Incorrect password"
Firebase error	<code>FirebaseAuthException</code> → displayed message
Firebase not configured	Automatic fallback to demo mode

File: `lib/services/auth_service.dart` → `AuthResult` class

Instructor mark: _____ / 3

Comments: _____

3. Implemented modules

Mapping to suggested CAT modules:

Suggested module	Nikah Planner module	File	Instructor mark
1. Login Screen	Sign-in screen	<code>lib/features/auth/login_screen.dart</code>	
2. Dashboard	Home (countdown, tasks, stats)	<code>lib/features/home/home_screen.dart</code>	
3. Student Registration	Couple / user registration	<code>lib/features/auth/register_screen.dart</code>	
4. Local Database	SharedPreferences + mock data	<code>local_wedding_store.dart</code> , <code>mock_data.dart</code>	

Suggested module	Nikah Planner module	File	Instructor mark
5. API Integration	Firebase Auth + Firestore	auth_service.dart , firestore_service.dart	
6. Reports Screen	Budget report (chart + transactions)	lib/features/budget/budget_screen.dart	

Additional screens: Guests, Gallery, Schedule, Vendors, Profile, Invitations, Notifications, Settings.

Module comments: _____

4. Grading (instructor)

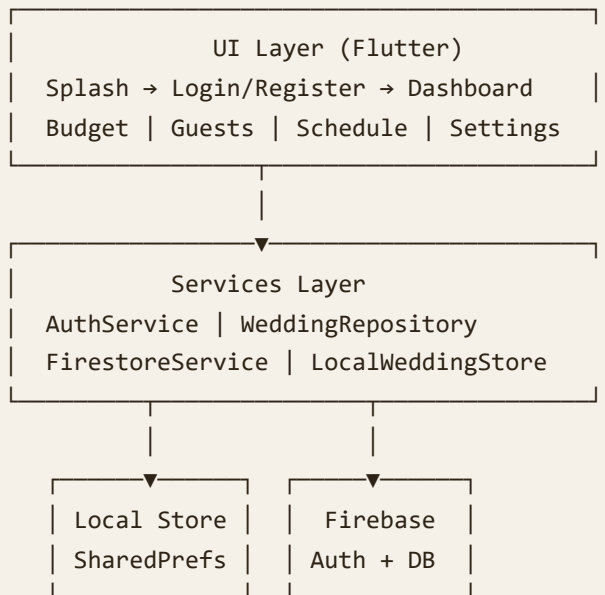
To be completed by the instructor.

Criterion	Max	Evidence in project	Mark
User Interface Design	5	13 screens, consistent theme, reusable widgets	
Navigation	4	GoRouter, 5 tabs, sub-pages	
Local Storage	5	SharedPreferences + mock_data	
Data Retrieval	4	MockData, WeddingRepository, filters	
Networking / API	5	Firebase Auth + Firestore	
Error Handling	3	AuthResult, UI error messages	
Code Quality	2	features/, services/ structure, Flutter lints	
Documentation	2	README, this logbook, SOUMISSION_PROF.md	
Total	30		

Instructor signature: _____

Date: _____

5. Technical architecture



Pattern: Repository pattern — `WeddingRepository` abstracts the data source (local or cloud).

6. Local storage and data retrieval

6.1 Embedded data (instructor demo mode)

File `mock_data.dart` — sample data for Siti & Ahmad to test without an account:

- 8 budget categories, 8 guests, 6 events, 7 vendors

6.2 Persistent storage (`SharedPreferences`)

Each registered user gets a private workspace:

- Registration → `WeddingDataFactory.createNew()` → JSON save
- Login → `loadForUser()` → personal data displayed

6.3 Retrieval example

```
// Simplified example
final profile = await WeddingRepository.instance.loadForUser(userId);
WeddingSession.activate(profile);
// Screens read MockData which points to WeddingSession
```

7. Networking and API consumption

7.1 Client-server architecture

```
[Flutter App] ↔ [Firebase Auth API]    (authentication)
[Flutter App] ↔ [Cloud Firestore API]  (wedding data)
```

7.2 Firestore collections

```
users/{userId}      → email, name, role, weddingId
weddings/{weddingId} → brideName, groomName, weddingDate, budget...
weddings/{id}/guests/ → guest list
```

7.3 Asynchronous processing

All network operations use `async/await` :

- `signIn()`, `register()`, `createWeddingProfile()`, `loadForUser()`
-

8. Error handling

```
// AuthResult pattern used across the app
final result = await authService.signIn(email: email, password: password);
if (result.isSuccess) {
  context.go('/home');
} else {
  setState(() => _error = result.error); // Display to user
}
```

Principles:

- Client-side validation before API calls
 - Catch `FirebaseAuthException` with readable messages
 - Graceful fallback when Firebase is unavailable (demo mode)
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9. Development phases (logbook)

Week	Date	Activity	Status	Instructor mark
1	[Date]	Requirements analysis, React UI mockup	Done	
2	[Date]	Flutter project setup, theme, Splash	Done	
3	[Date]	Login, Register, GoRouter navigation	Done	
4	[Date]	Dashboard, Budget, Guests, Schedule	Done	
5	[Date]	Local storage with SharedPreferences	Done	
5	[Date]	Firebase integration (Auth + Firestore)	Done	
6	[Date]	Android testing, APK build, GitHub publish	Done	
6	[Date]	Documentation and logbook	Done	

Logbook comments: _____

10. Testing and demonstration instructions

10.1 Mobile application (instructor)

```
git clone https://github.com/moinaalia/nikah-planner.git
cd nikah-planner/nikah_planner
flutter pub get
flutter run
```

Recommended demo flow:

1. Splash screen → Login
2. Demo sign-in (any email + password after registration, or demo data)
3. Dashboard: countdown, tasks, budget stats
4. Budget: pie chart + transactions (Reports)
5. Guests: RSVP filters
6. Schedule: event timeline
7. Settings → Sign Out

10.2 Web prototype (browser)

```
cd nikah-planner
npm install
npm run dev
# Open http://localhost:5173
```

10.3 Personal account (registration)

1. Register → fill form → Create Account
2. Each email = separate private workspace
3. Sign Out → create a second account → show different data

11. Required screenshots

Included in the `screenshots/` folder:

#	Screen	File
1	Splash	<code>01_splash.png</code>
2	Login	<code>02_login.png</code>
3	Register	<code>03_register.png</code>
4	Dashboard (Home)	<code>04_dashboard.png</code>
5	Budget / Reports	<code>05_budget.png</code>
6	Guest List + filters	<code>06_guests.png</code>
7	Schedule	<code>07_schedule.png</code>
8	Settings + profile	<code>08_settings.png</code>
9	Login error message	<code>09_error.png</code>

How to capture: Android emulator camera button, or `flutter run` on a phone → screenshot.

12. Revision questions

12.1 Temporary vs permanent storage

Temporary	Permanent
RAM / in-memory session (<code>WeddingSession</code>)	SharedPreferences on disk
Lost when app closes	Persists after restart
<code>setState</code> variables	JSON file in SharedPreferences
Example: active guest filter	Example: saved user account

12.2 Explain SQLite

SQLite is a lightweight relational database embedded on the mobile device. It stores data in tables using SQL (SELECT, INSERT, UPDATE). It is an alternative to SharedPreferences for complex structured data. *Nikah Planner uses SharedPreferences (key-value JSON); SQLite could be added for large guest lists.*

12.3 Define API

An **API** (Application Programming Interface) is an interface that allows two programs to communicate. On mobile, the app (client) sends HTTP requests to a server (e.g. Firebase) and receives JSON responses.

12.4 Explain asynchronous processing

Asynchronous processing lets the app keep running during long operations (network, disk). In Dart: `async/await` and `Future`. Example: `await authService.signIn()` does not freeze the UI.

12.5 Compare GET and POST

GET	POST
Retrieve data	Send / create data
Parameters in URL	Data in request body
Example: read Firestore profile	Example: create Firebase Auth account
Idempotent	May modify the server

12.6 Client-server architecture

The **client** (Flutter mobile app) sends requests to the **server** (Firebase Cloud). The server processes them, accesses the database, and returns a response. The app never accesses the server database directly without going through the API.

12.7 Explain JSON

JSON (JavaScript Object Notation) is a text format for structured data: `{ "brideName": "Siti", "budget": 15000 }`. Used by Firestore and SharedPreferences in Nikah Planner.

12.8 Design an app with storage and networking

Nikah Planner illustrates this design:

1. **Registration** → POST Firebase Auth + local save
 2. **Read** → GET Firestore or SharedPreferences
 3. **Display** → Flutter UI reads `MockData` / `WeddingSession`
 4. **Offline** → local storage fallback when no network
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13. References

1. Android Developer Documentation — <https://developer.android.com>
 2. Flutter Documentation — <https://docs.flutter.dev>
 3. Firebase Documentation — <https://firebase.google.com/docs>
 4. SQLite Documentation — <https://www.sqlite.org/docs.html>
 5. REST API Design — <https://restfulapi.net>
 6. SharedPreferences (Flutter) — package `shared_preferences`
 7. Project repository — <https://github.com/moinaalia/nikah-planner>
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14. Appendices

A. Submission links

Item	Link
Source code	https://github.com/moinaalia/nikah-planner
GitHub profile	https://github.com/moinaalia
README	https://github.com/moinaalia/nikah-planner#readme
Logbook PDF	https://github.com/moinaalia/nikah-planner/raw/main/JOURNAL_DE_BORD.pdf

B. CAT submission checklist

- ☒ Source code on GitHub
- ☒ Screenshots (section 11)
- ☒ Technical report (PDF)
- ☐ Live demonstration (emulator or phone)

C. Repository structure

```
nikah-planner/  
├─ nikah_planner/    ← Flutter app (main CAT deliverable)  
├─ screenshots/      ← UI screenshots  
├─ src/              ← React prototype  
├─ README.md  
├─ SOUMISSION_PROF.md  
├─ JOURNAL_DE_BORD.md ← This document  
└─ JOURNAL_DE_BORD.pdf
```

Report prepared in accordance with CAT instructions — Mobile Application Development.